

RaceDay API Endpoint Plan - Model Answer

Part 1 planning only: this document defines the proposed RESTful API; it does not implement controllers or the Web API.

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new RaceDay user and creates the appropriate Organiser or Participant profile.	Public	{ fullName, email, password, role, profileDetails }	201 Created; 400 Bad Request; 409 Conflict
POST	/api/auth/login	Authenticates a registered user and returns user/role information and an access token.	Public	{ email, password }	200 OK; 400 Bad Request; 401 Unauthorized

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the profile of the currently authenticated RaceDay user.	Any authenticated user	None	200 OK; 401 Unauthorized; 404 Not Found
PUT	/api/users/me	Updates editable details for the currently authenticated user and their role-specific profile.	Any authenticated user	{ fullName, contactNumber?, emergencyContact? }	200 OK; 400 Bad Request; 401 Unauthorized

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Returns all RaceDay events available for browsing.	Public	None	200 OK
GET	/api/events/{id}	Returns the details of one RaceDay event using the supplied event ID.	Public	None	200 OK; 404 Not Found
POST	/api/events	Creates a new RaceDay event owned by the authenticated Organiser.	Organiser	{ eventName, description, eventDate, location, distanceKm, eventType, maxParticipants }	201 Created; 400 Bad Request; 401 Unauthorized; 403 Forbidden
PUT	/api/events/{id}	Updates an existing event owned by the authenticated Organiser.	Organiser	{ eventName, description, eventDate, location, distanceKm, eventType, maxParticipants, status }	200 OK; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found
DELETE	/api/events/{id}	Deletes an event owned by the authenticated Organiser when business rules allow deletion.	Organiser	None	204 No Content; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Returns all categories configured for a specific event.	Public	None	200 OK; 404 Not Found
POST	/api/events/{eventId}/categories	Creates a category for an event managed by the authenticated Organiser.	Organiser	{ categoryName, minAge, maxAge, fee }	201 Created; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found
PUT	/api/events/{eventId}/categories/{categoryId}	Updates an existing category for an Organiser-owned event.	Organiser	{ categoryName, minAge, maxAge, fee }	200 OK; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found
DELETE	/api/events/{eventId}/categories/{categoryId}	Deletes an event category when it is not protected by existing enrolments.	Organiser	None	204 No Content; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Enrols the authenticated Participant in an event and records the selected category.	Participant	{ categoryId }	201 Created; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/enrolments/me	Returns all event enrolments belonging to the authenticated Participant.	Participant	None	200 OK; 401 Unauthorized; 403 Forbidden
DELETE	/api/enrolments/{id}	Cancels an enrolment belonging to the authenticated Participant when cancellation is permitted.	Participant	None	204 No Content; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict
GET	/api/events/{eventId}/enrolments	Returns enrolments for an event owned by the authenticated Organiser.	Organiser	None	200 OK; 401 Unauthorized; 403 Forbidden; 404 Not Found

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Captures finish time and finishing position for an enrolment in an Organiser-owned event.	Organiser	{ finishTime, position }	201 Created; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict
PUT	/api/enrolments/{enrolmentId}/result	Corrects or updates a previously captured result for an Organiser-owned event.	Organiser	{ finishTime, position }	200 OK; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found
GET	/api/results/me	Returns race results for the authenticated Participant across completed enrolments.	Participant	None	200 OK; 401 Unauthorized; 403 Forbidden
GET	/api/events/{eventId}/results	Returns the recorded results for a specific event for management/reporting.	Organiser	None	200 OK; 401 Unauthorized; 403 Forbidden; 404 Not Found

API Planning Notes

Role values: Public = no authentication required; Any authenticated user = either role; Organiser = Organiser only; Participant = Participant only.

Status codes: 200 for successful reads/updates, 201 for newly created resources, 204 for successful deletes, 400 for invalid input, 401 for missing/invalid authentication, 403 for valid user without permission, 404 when the resource does not exist, and 409 when the request conflicts with current data or state.

Ownership checks: Organiser-only update/delete/result actions should verify that the authenticated Organiser owns the relevant Event.

Enrolment rule: the Participant selects a Category belonging to the target Event; duplicate Participant/Event enrolments should return 409 Conflict.

Consistency with the database: the plan maps to Users, Organisers, Participants, Events, Categories, Enrolments and Results in the ERD/SQL model.